

CLASS -10 (2025-26)

Elementary Concept of Objects and Class

CHAPTER 2

Assignments:-

A. Tick (✓) the correct answer

Question 1. If `Animal` is a class, then which of the following can be the object of the `Animal` class?

- a. Tiger
- b. Deer
- c. Lion
- d. All of these

Ans- d. All of these

Question 2. Which of the following can be defined as a user-defined data type?

- a. Class
- b. Object
- c. Primitive
- d. None of these

Ans- a. Class

Question 3. Which of the following keywords are used to define a class?

- a. `class`
- b. `public`
- c. Both a and b
- d. None of these

Ans- c. Both a and b

Question 4. Which of the following is the correct way to create an object of the class "Smartphone"?

- a. `Smartphone obj = new Smartphone;`
- b. `Smartphone obj = new Smartp`
- c. `Smartphone object = Smartphone();`
- d. None of these

Ans- d. None of these

Correct syntax: `Smartphone obj = new Smartphone();`

Question 5. Which of the following represents the constructor of the `ABC` class?

- a. `ABC()`
- b. `ABC`
- c. `ABCConstructor`
- d. None of these

Ans- a. `ABC()`

B. Short Answer Type Questions

Question 1. What is a class?

Ans- A class is a blueprint or template for creating objects. It defines properties (variables) and behaviors (methods) of an object.

Example:

```
class Animal {  
    String name;  
    void sound() {  
        System.out.println("Animal makes a sound");  
    }  
}
```



Question 2. What is an object?

Ans- An object is an instance of a class. It represents a real-world entity that has state and behavior.

Example:

```
Animal tiger = new Animal();
```

Question 3. What is the process of creating instances of a class called?

Ans- The process of creating instances of a class is called **Instantiation**. It is done using the **new** keyword.

Example:

```
Smartphone myPhone = new Smartphone();
```

Extra Questions – Chapter 2: Classes and Objects

1. Multiple Choice Questions

Q1. What keyword is used to create an object in Java?

- a) object
- b) create
- c) new
- d) instance

Ans: c) new

Q2. Which of the following is a special method used to initialize objects?

- a) Setter
- b) Constructor
- c) Getter
- d) Objectifier

Ans: b) Constructor

2. Fill in the Blanks

Q3. A _____ is a blueprint for creating objects.

Ans: class

Q4. The process of creating an object from a class is called _____.

Ans: instantiation

Q5. A constructor must have the same _____ as the class.

Ans: name

3. True or False

Q6. Every Java program must contain at least one class.

Ans: True

Q7. The `main()` method is required to run a Java program.

Ans: True

4. Short Answer Questions

Q8. What is a constructor?

Ans: A constructor is a special method that is automatically called when an object is created. It is used to initialize objects.

Q9. Write the syntax to define a class named `Book`.

Ans:

```
class Book {  
    // fields and methods  
}
```

Q10. Create an object of a class named `Laptop`.

Ans:

```
Laptop myLaptop = new Laptop();
```



Assertion and Reasoning Questions – Chapter 2: Classes and Objects

Q1.

Assertion (A): A class is a user-defined data type.

Reason (R): It defines variables and methods under a single unit.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: a

Q2.

Assertion (A): Objects are created using the `new` keyword.

Reason (R): The `new` keyword allocates memory for the object.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: a

Q3.

Assertion (A): Constructors can have a different name than the class.

Reason (R): Constructors are used to destroy objects.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: d

Q4.

Assertion (A): An object is an instance of a class.

Reason (R): Objects cannot access class methods.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: c

Q5.

Assertion (A): A constructor can be overloaded.

Reason (R): Java allows multiple constructors with different parameter lists.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: a



Q6.

Assertion (A): The keyword `class` is used to define a class in Java.

Reason (R): Java does not support user-defined data types.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: c

Q7.

Assertion (A): `Smartphone phone = new Smartphone();` creates a new object.

Reason (R): This syntax calls the class's constructor.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: a

Q8.

Assertion (A): An object must be declared before it is used.

Reason (R): Declaration tells the compiler about the object type and name.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: a

Q9.

Assertion (A): A class cannot contain methods.

Reason (R): Only variables are allowed in a class.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: d

Q10.

Assertion (A): A constructor is automatically called when an object is created.

Reason (R): It initializes the state of the object.

Options:

- a) Both A and R are true and R is the correct explanation of A
- b) Both A and R are true but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer: a



One-Word Questions and Answers

1. Q: What keyword is used to define a class in Java?
A: **class**
2. Q: What keyword is used to create an object in Java?
A: **new**
3. Q: What is the process of creating an object called?
A: **Instantiation**
4. Q: Which method initializes objects in Java?
A: **Constructor**
5. Q: What is an instance of a class called?
A: **Object**
6. Q: Which user-defined data type is used to create objects?
A: **Class**
7. Q: What is the blueprint of an object in Java?
A: **Class**
8. Q: What keyword is used to make class members accessible outside the class?
A: **public**
9. Q: What is the name of the default method that runs a Java program?
A: **main**
10. Q: What is automatically called when an object is created?
A: **Constructor**

Fill in the Blanks with Answers

1. A _____ is a blueprint for creating objects.
Answer: **class**
2. An _____ is an instance of a class.
Answer: **object**
3. The keyword used to create an object is _____.
Answer: **new**
4. A _____ method is used to initialize an object.
Answer: **constructor**
5. The process of creating an object is called _____.
Answer: **instantiation**
6. A constructor must have the same _____ as the class.
Answer: **name**
7. The keyword _____ is used to define a class in Java.
Answer: **class**
8. Java allows multiple constructors in a class. This feature is called constructor _____.
Answer: **overloading**
9. The _____ method is the entry point of a Java program.
Answer: **main**
10. A class can contain both _____ and _____.
Answer: **variables, methods**

Short Answer Questions with Answers



1. **Q1. What is a class in Java?**

Ans: A class is a user-defined blueprint or prototype from which objects are created. It contains fields (variables) and methods to define behaviors.

2. **Q2. What is an object in Java?**

Ans: An object is an instance of a class. It has its own identity, state, and behavior defined by the class.

3. **Q3. How is an object created in Java?**

Ans: An object is created using the `new` keyword followed by the class constructor.

Example: `Student s = new Student();`

4. **Q4. What is a constructor?**

Ans: A constructor is a special method that is automatically called when an object is created. It is used to initialize the object.

5. **Q5. Can a constructor be overloaded?**

Ans: Yes, constructors can be overloaded by defining multiple constructors with different parameter lists in the same class.

6. **Q6. What is instantiation?**

Ans: Instantiation is the process of creating an object from a class using the `new` keyword.

7. **Q7. What is the syntax to define a class in Java?**

Ans:

```
class ClassName {  
    // fields and methods  
}
```

8. **Q8. What is the role of the `main()` method in Java?**

Ans: The `main()` method is the entry point of any Java program. It tells the JVM where to start execution.

9. **Q9. Can a class have more than one object?**

Ans: Yes, a class can have multiple objects, each with its own data and state.

10. **Q10. What is the difference between a class and an object?**

Ans: A class is a template or blueprint, whereas an object is an instance of the class created during runtime.

